What a Fracture Detector Really Learns: A Leak-Controlled Audit

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Abstract

We train a commodity fracture classifier on public radiographs and audit what it has learned: leakage, failure modes, shortcut cues and external transfer. A ConvNeXt-Tiny trained on FracAtlas with a leak-controlled five-fold split reaches a pooled out-of-fold AUC of 0.923 (95% CI 0.910 to 0.935) on threshold-shifted scores. Scored without retraining on GRAZPEDWRI-DX (20,327 paediatric wrist images, patient-grouped), AUC falls to 0.825, and to 0.697 on initial examinations. The model responds strongly to cast and treatment status: within the same patient, label and view, a cast adds about 4 logits to the score, roughly twice the fracture effect at first presentation, and the model's own score separates cast from no cast (AUC 0.932) better than fracture from no fracture (0.825). Feature probes, Grad-CAM, perturbations and a single-reader visual review of the training data, done by an AI model and not a radiologist (casts in 16 of 60 random fractured versus 1 of 60 non-fractured FracAtlas images) point the same way; the evidence is suggestive, not causal. We then try to reduce misses of the smallest fractures (sensitivity 0.761 on the smallest quartile) with box-supervised auxiliary heads, a finer localisation head and oversampling. The outcome is *inconclusive*: the unmodified model moves by +0.033 in small-fracture sensitivity between two random seeds, and no box-supervised run beats its no-box control. Auxiliary-head variants gain +0.033 to +0.072 against the seed-42 baseline, 0.000 to +0.039 against the seed-43 baseline, and +0.017 to +0.056 against the mean of the two; resolving gains of this size needs several seeds per arm, which we did not run. An illustrative emergency-department simulation shows how external performance changes the alert burden. All results use public data only; this is not a clinical validation.

1 Introduction

Reported accuracy of fracture-detection models is usually internal, and leakage, shortcut cues and distribution shift are rarely audited together. We ask three questions with a single reproducible pipeline: (i) how accurate is a commodity classifier under a leak-controlled split; (ii) does it transfer to an unseen dataset, and what does it respond to there; (iii) can a localisation-based modification reduce misses of small fractures, and can we tell if it does? The contribution is the evaluation, including its negative findings, not a new architecture.

2 Data

FracAtlas (development). FracAtlas [1] contains mixed-body-part radiographs with fracture labels and bounding boxes for fractured images. There are no patient identifiers, so we form groups from perceptual-hash near-duplicates plus consecutive multi-scan runs, and split by

group. Subgroups with fewer than 30 positives or negatives (for example hip and shoulder, 10 positives each) are flagged unreliable. There is no cast or treatment label; the `hardware` field marks internal implants only.

GRAZPEDWRI-DX (external). GRAZPEDWRI-DX [2] has 20,327 paediatric wrist images, grouped by patient. It is used for scoring only, with the threshold fixed beforehand. The `device_manufacturer` field lists 20,215 Siemens and 112 Agfa images, so a device effect can be tested only on 112 images (54 patients). The dataset also records cast, metal, projection, initial examination versus follow-up, age and sex. Dataset licences were not verified at the time of writing (see `docs/DATA_LICENCES.md`).

3 Methods

Model and training. ConvNeXt-Tiny [3] (timm), single channel, 512 px, AdamW with one-cycle schedule, mixed precision, batch 16, up to 12 epochs with patience 4. Five-fold out-of-fold evaluation: test fold k, validation fold $(k + 1) \bmod 5$.

Operating point. For each fold model the threshold is the 90th percentile of its validation-fold negative scores, which gives validation specificity of 0.899 to 0.900. Each fold's scores are shifted so that zero is its operating point, and sensitivity and specificity are reported at that fixed rule.

External scoring. Each of the five fold models scores every GRAZ image, its logit is shifted by that model's own validation threshold, and the external score is the mean of the five shifted logits, flagged if at least zero. External results therefore describe this five-model average, not a single model.

Uncertainty. 95% group-bootstrap confidence intervals (whole groups or patients resampled) with 1,000 draws, except the resolution test (300 draws) and the feature-probe AUCs (200 draws); paired bootstrap for differences between runs on identical folds. The view-combination and AO-class intervals were recomputed with 1,000 draws; no value moved by more than 0.004.

Small-fracture subset. Positives whose box area relative to the image is at or below the 25th percentile (180 images, about 35 per fold). The primary metric is sensitivity there at the fixed threshold.

Box-supervision experiments. An auxiliary head predicts a map of box locations during training only (removed before saving). The coarse head gives a 16×16 map (32 px cells); a fine head uses a feature-pyramid design for a 32×32 map (16 px cells). Targets are max-pooled box masks (boxes grown by 16 px), positive weight 10, with small boxes upweighted. Boxes are used for 100%, 25% or 10% of fractured training images. Controls: the same head supervised on negatives only (no box information), and the plain model with a second seed. A further candidate oversamples smallest-quartile positives ×3.

Analysis plan. The decision rule (small-fracture sensitivity CI lower bound > 0 versus *both* baseline seeds, AUC not worse than -0.01, specificity not worse than -0.03, then replication and an external check) is a self-timestamped local file (`docs/OVERNIGHT_PROTOCOL.md`), not an external registration. The box 100/25/10% runs and the no-box auxiliary control had been

seen before it was written and are exploratory; only the fine head, its control and oversampling were run afterwards. The rule was tightened once, after the seed-43 baseline appeared. The comparison with the mean of the two baselines was added after seeing the results.

4 Results

4.1 Internal performance

The headline AUC of 0.923 (0.910 to 0.935) is computed on the pooled out-of-fold scores after each fold's threshold shift. Pooling the unshifted logits gives 0.911, and the mean of the five per-fold AUCs is 0.931; we report the shifted pooled value because the shared threshold is how the model would be used. At that threshold sensitivity is 0.808 and specificity 0.897. Excluding hardware-flagged images gives 0.915. The hardware flag alone predicts with AUC 0.567 and image-size class with 0.536. Sensitivity on the smallest-fracture quartile is lower than on the largest (72.2% vs 87.2%; here "small" is defined by the box's longer side, whereas Section 4.5 uses relative box area, which gives 0.761 for the same baseline). Naive and group-controlled splits give 0.931 and 0.923 with overlapping intervals, so leakage inflation is *not demonstrated*. Grad-CAM [4] peaks fall inside the annotated box for 63.7% (59.6 to 67.8) of detected fractures and 29.7% of missed ones (chance about 3%). Occluding the fracture lowers the score by 2.42 logits (2.13 to 2.72), whereas a same-size control patch elsewhere slightly raises it (by about 0.01 logits on average); 86% of images stay flagged after occluding the fracture. Saliency supports review, not mechanism.

4.2 External transfer to GRAZPEDWRI-DX

Subset	AUC	Sensitivity	Specificity
FracAtlas hand, internal	0.899 (0.876 to 0.920)	0.866	0.742
GRAZ, all	0.825 (0.816 to 0.833)	0.776	0.749
GRAZ, no cast, no metal*	0.744 (0.735 to 0.753)	0.61	0.79
GRAZ, initial exams only*	0.697 (0.687 to 0.707)	0.540	0.790
GRAZ, cast present (96% fractured)	0.560	0.995	0.017

Table 1: Internal and external performance at the pre-specified threshold. Prevalence differs (17.6% vs 67%), so only AUC, sensitivity and specificity are comparable. *Subsets chosen after the overall external results were seen.

Casted images are flagged almost regardless of label (Section 4.3). Frontal views are weakest (initial exams: AUC 0.61, sensitivity 31%, versus lateral 0.79 and 73%); combining views by maximum or mean did not help (first-visit studies with both views, n=4951: frontal 0.580, lateral 0.785, max 0.775, mean 0.743). Sensitivity varies by AO class (51% for 23r-M, the most common, to 79% for 22r-D). Re-fitting the threshold on GRAZ gives sensitivity 0.673 at 90% specificity, so the ranking, not only the threshold, transfers poorly.

4.3 Shortcut audit

In GRAZ, casts occur almost only at follow-up (2 casted images among 10,861 initial-exam images), and un-casted follow-ups already score high, so a pooled cast comparison mixes cast with treatment stage. We therefore compared images within patients and within follow-ups, and added probes, saliency, perturbations and a visual review of the training data (Table 2). All

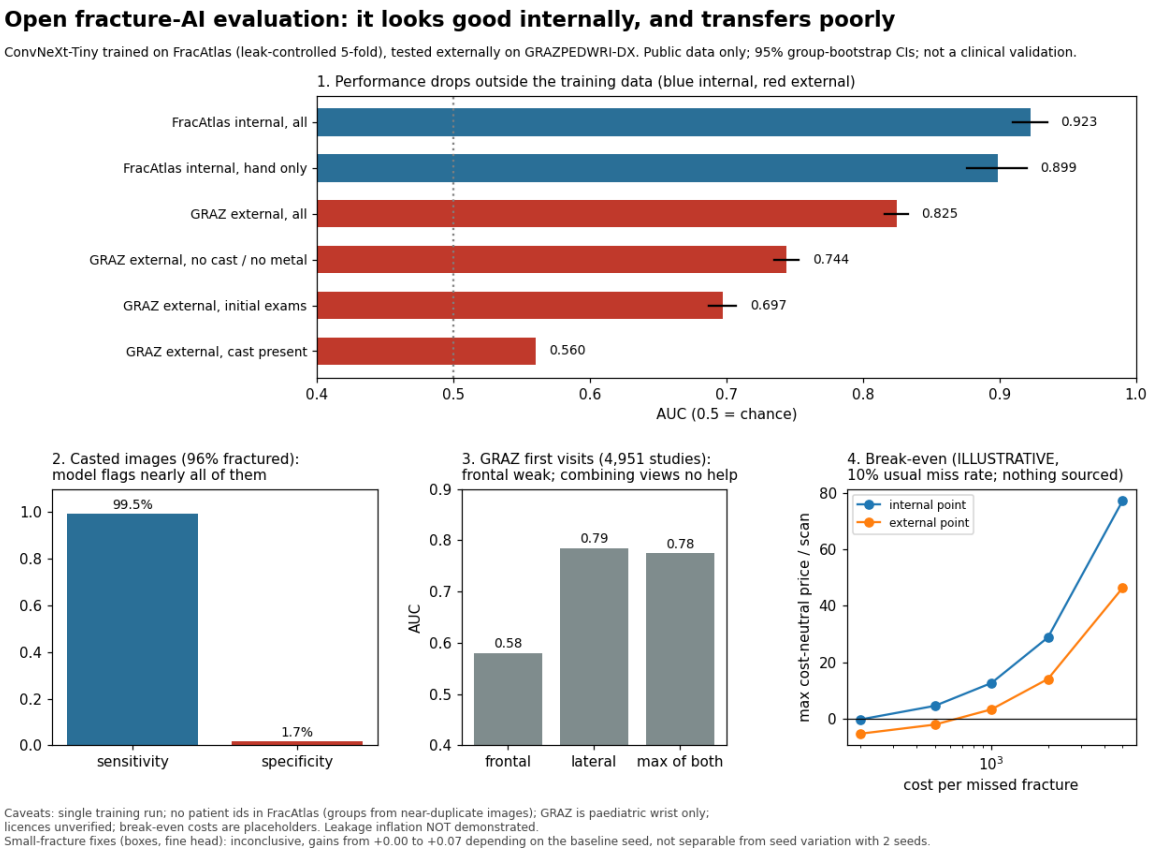


Figure 1: One-page summary of the audit (AUC by subset, behaviour on casted images, view effect, illustrative break-even). The small-fracture range in the figure footer is superseded by Section 4.5, which names the baseline each range is measured against.

analyses are post hoc and use the five-model external score; nothing was retrained apart from linear probes.

Cast and treatment status. Within the same patient, label and view, a cast adds about 4 logits, roughly twice the fracture effect at first presentation, and the same shift appears in non-fractured follow-ups (only 10 patients give a paired non-fractured contrast, so that estimate is imprecise). The earlier comparison of a 7.7-logit cast rise against a 2.8-logit fracture effect was confounded, because all 237 casted non-fractured images are follow-ups; it is replaced by the analyses above. Cast status is almost perfectly linearly decodable from the features, though a probe shows only that the information is present (raw pixels also give 0.935). On casted wrists Grad-CAM moves off bone towards cast, soft-tissue and air intensities. Blur and border masking lower casted scores two to four times more than un-casted ones, which points to fine texture and cast edges, while masking the brightest pixels does not lower casted scores; these perturbations also move images off the training distribution. In FracAtlas, casts and splints accompany fractures (about one in four random fractured images) and are almost absent otherwise, a plausible source for this behaviour, in line with shortcut learning [5] reported for chest radiographs [6]. The visual review was done by one non-radiologist reader (an AI model) on 512 px images without a second reader. Taken together, the model responds to cast and treatment status; the evidence is suggestive, not causal, and within patients the casted exam was the earlier one in 96% of pairs, so cast remains partly confounded with time since injury.

Analysis	Result
Cast effect within patient, same label and view, fractured follow-up wrists (946 patients)	+4.16 logits (3.99 to 4.31); larger on frontal (+5.68) than lateral (+2.61)
Cast effect within follow-ups, unpaired, non-fractured (244 vs 237 images)	+3.84 (2.71 to 5.13); flag rate 59% to 98%
Same, fractured (3,448 vs 5,537 images)	+4.21 (4.04 to 4.37); flag rate 74% to 99.5%
Fracture effect at initial exams without cast	about +2.1 (mean −1.16 vs +0.93)
Model's own score as a classifier (all 20,327 images)	cast vs no cast AUC 0.932 (0.927 to 0.937); fracture AUC 0.825
Linear probe on pooled features (3,967 images)	cast 0.998, follow-up 0.960, fracture 0.907 (raw-pixel probe for cast 0.935)
Grad-CAM, share of top-10% saliency on bright (bone-like) pixels	0.356 un-casted to 0.226 casted (difference −0.130, −0.159 to −0.103)
Gaussian blur (σ=4); 10% border masked	casted −5.56 vs un-casted −1.34 logits; −1.98 vs −0.67
Masking brightest 5% of pixels	raises scores in both groups (casted +1.42, un-casted +4.60; shifted control +1.08, +2.14)
FracAtlas visual review, cast or splint	16/60 random fractured vs 1/60 random non-fractured ($p < 0.001$); 5/60 of top-scoring false alarms
Device, all images	Agfa (112) AUC 0.774 (0.640 to 0.892) vs Siemens 0.825 (0.817 to 0.833)
Age and sex, no-cast initial exams	AUC 0.645 to 0.72 in every subgroup; age 16+ weakest (AUC 0.645, sensitivity 0.45)

Table 2: Shortcut audit on GRAZPEDWRI-DX and FracAtlas. Score changes are in logits of the five-model external score; intervals are 95% patient-bootstrap.

Device, age and sex. Agfa and Siemens intervals overlap, and the within-patient Agfa minus Siemens difference is −0.40 logits (−1.32 to +0.48, 35 patients); 112 images can rule out only very large device effects. On all images, age and sex differences partly track case mix. Restricted to no-cast initial exams, AUC lies between 0.645 and 0.72 in every age band and both sexes; the 16+ band is weakest (AUC 0.645, sensitivity 0.45), sensitivity falls steadily with age at the fixed threshold, and males score modestly above females (0.717 vs 0.675).

4.4 Resolution is not the bottleneck

Downscaling to 384/256/192/128 px then restoring 512 px gives AUC 0.919/0.909/0.894/0.850 (native 0.923); sensitivity stays near 0.80 and specificity falls, so blur adds false alarms rather than misses. Upscaling inputs to 640/768 px gives no gain. Small-fracture sensitivity does not fall with blur. This tests a model trained at 512 px only.

4.5 Can missed small fractures be fixed?

Against seed 42 alone, three runs would be called wins (boxes 100%, boxes 10%, fine head with boxes). But the unmodified model moved +0.033 between seeds, and no box run, compared with its no-box control or with the better baseline (seed 43), has a confidence interval above zero (fine boxes vs fine control: +0.033, −0.010 to +0.076). Oversampling did not help. The gains therefore depend on the comparison: for the six auxiliary-head variants they are +0.033 to +0.072 against seed 42, 0.000 to +0.039 against seed 43, and +0.017 to +0.056 against the mean of the two baselines (across all seven candidates, including oversampling, −0.017 to +0.056). Against that mean, three box runs have intervals above zero; the strict rule leans towards the

Run	AUC	Small sens.	vs seed 42	vs seed 43	vs mean of both
Baseline, seed 42	0.923	0.761	reference		
Baseline, seed 43	0.932	0.794	+0.033 (−0.010, +0.079)	reference	
Aux head, no boxes (control)	0.935	0.806	+0.044 (+0.000, +0.090)	+0.011 (−0.033, +0.052)	+0.028 (−0.010, +0.066)
Boxes 100%	0.936	0.833	+0.072 (+0.034, +0.119)	+0.039 (−0.005, +0.078)	+0.056 (+0.021, +0.090)
Boxes 25%	0.932	0.800	+0.039 (−0.015, +0.097)	+0.006 (−0.043, +0.051)	+0.022 (−0.024, +0.070)
Boxes 10%	0.938	0.828	+0.067 (+0.019, +0.120)	+0.033 (−0.018, +0.078)	+0.050 (+0.006, +0.093)
Fine head, boxes 100%	0.939	0.828	+0.067 (+0.011, +0.126)	+0.033 (−0.013, +0.080)	+0.050 (+0.003, +0.097)
Fine head, no boxes (control)	0.936	0.794	+0.033 (−0.017, +0.086)	+0.000 (−0.045, +0.045)	+0.017 (−0.025, +0.059)
Oversample small ×3	0.932	0.761	+0.000 (−0.053, +0.052)	−0.033 (−0.086, +0.015)	−0.017 (−0.064, +0.029)

Table 3: Small-fracture sensitivity at the fixed 90%-specificity threshold (180 positives) and its change against each baseline. Paired group-bootstrap 95% intervals in brackets. The comparison with the mean of both baselines is exploratory and post hoc.

null and the pooled comparison against it. The seed yardstick is a single pair of runs, so it shows seed variation matters but does not measure it. Any auxiliary head raised overall AUC by about 0.01 with or without boxes, close to the seed difference of +0.008. No candidate met the pre-set rule, so replication and the external check were not triggered. The paired 95% half-width on these 180 images is about 0.045, so differences of about 0.065 or more are detectable in principle, before run-to-run variance; gains of the size seen here are not separable from noise with two seeds. A decisive test needs three to five seeds per arm, which were not run.

4.6 Illustrative break-even

Treating the model as a safety-net second reader with independence assumed (optimistic), the maximum cost-neutral price per scan is lower at the external initial-exam operating point than the internal one in every cell of a grid of placeholder costs and usual miss rates (for example 3.2 vs 12.6 at a 10% miss rate and 1,000 per miss) and negative in more cells (11 of 25 external, 5 internal). Costs are not sourced; the point is only that external, not internal, performance decides viability.

4.7 Illustrative emergency-department simulation

We built a site-parameterised discrete-event simulation of an ED fracture pathway, implemented in `fracai/twin.py`: limb X-rays arrive with a daytime-peaked pattern, a single reporter works 9 hours a day, and the model acts either as an alert at the time of the X-ray (clinician re-reads flagged images; catch probability 0.8, independence assumed, which is optimistic) or as a triage of the reporting queue. Volume is derived from NHS England's 1.503 million Type 1 attendances in September 2026; everything else is assumed and labelled so: 120 departments, 12% limb X-ray share (about 49 a day), 20% fracture prevalence, a 5% usual miss rate (published studies give roughly 1% to 4%), and the model's measured operating points (internal; external first-visit with the external sensitivity on missed fractures scaled by the internal ratio, an assumption). At 5% miss rate (about 182 missed fractures a year) and a reporter 75%/95% busy, an internal-point alert catches about 109 misses before the patient leaves and cuts undetected fracture-hours from 976/1760 to 382/722 a year, at about 12 extra re-reads a day (4 false alarms). The external point catches about 71 and gives 13.5 re-reads a day with 8.2 false alarms. Triage matters most when the reporter is busy. False alerts' harm, costs and patient harm from delay are not modelled. This is an illustration, not a prediction for any hospital. The code includes `calibrate` and `validate` steps; on synthetic data generated by the same code, calibration recovered the hidden site and validation passed, which is circular and shows only that the fitting works on a correctly specified model. It would become a digital twin of a real site only after calibration and validation on that site's data.

5 Related work in brief

Published small-fracture approaches include a two-stage scaphoid network that re-examines first-stage negatives [7], YOLOv8 detectors on GRAZPEDWRI-DX [8] (attention variants of YOLO on the same dataset report gains of about 1 to 3 mAP50 points, roughly 64% to 67%, in papers we did not cite individually), and an object-detection study on paediatric wrist fractures [9]. We have not covered the commercial AI fracture-detection literature, NICE's early value assessment of AI for fracture detection in urgent care, hidden-stratification work, or reporting guidelines (TRIPOD+AI, CLAIM, DECIDE-AI), which a full version should. Differences of that size are comparable to the seed noise we measured; we did not see seed repeats in those reports, but we did not review them exhaustively. This is a brief pointer, not a systematic review.

6 Limitations

Single training run per setting, with one extra seed only for the baseline, so run-to-run variance is not estimated. The comparison with the mean of the two baselines is post hoc. The GRAZ overall AUC (0.825) is inflated by the model's response to casts; the no-cast, no-metal (0.744) and initial-exam (0.697) subsets are fairer but were chosen after the overall results were seen, and the adult mixed-site to paediatric wrist transfer is partly a task change, not only a domain shift. The shortcut audit is post hoc and observational: no image was seen both with and without a cast, the within-patient contrast is still confounded with time since injury, the non-fractured paired contrast rests on 10 patients, and the FracAtlas visual review had a single non-radiologist reader. The device comparison rests on 112 Agfa images. The ED simulation inputs other than national volume are assumptions. FracAtlas has no patient identifiers, so residual leakage is possible. GRAZ is paediatric wrist only and its `fracture_visible` label provenance was not checked. Saliency is not evidence of mechanism. Training at higher resolution, a stricter same-bone occlusion control, and training with GRAZ patients included were not tested. Break-even inputs are placeholders. Dataset licences are unverified.

7 Conclusion

A commodity classifier that looks strong internally (pooled AUC 0.923) transfers poorly (0.825 overall, 0.744 without cast or metal, 0.697 on first visits) and responds strongly to cast and treatment status, an effect that survives a within-patient comparison but is not proven causal. Whether box supervision, a finer head or oversampling reduce missed small fractures is *inconclusive*: no variant beats its no-box control, and the auxiliary-head gains (0.000 to +0.039 against the stronger seed-43 baseline, +0.033 to +0.072 against seed 42) cannot be told from run-to-run variation with two baseline seeds. The pipeline's controls stopped three apparent wins against one baseline seed from being reported as findings. The next steps are three to five seeds per arm analysed jointly, a larger small-fracture test set, matched cast and no-cast images to test the cast response directly, and site data to calibrate the simulation.

Reproducibility

Code, scripts, fixed split files and the analysis plan are in the project repository (`README.md` gives the run order). All runs are resumable and offline.

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